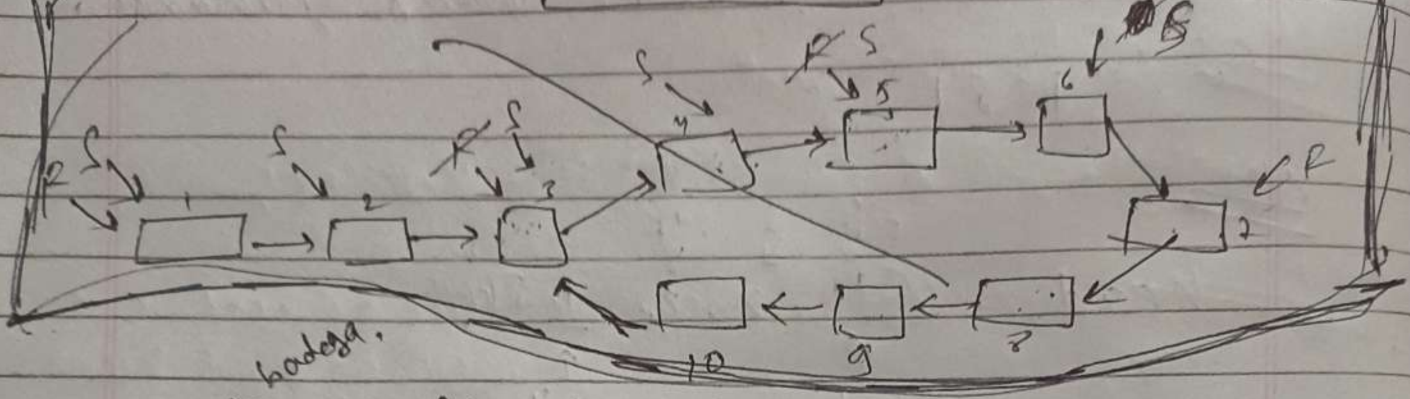
Find cycle in the LL

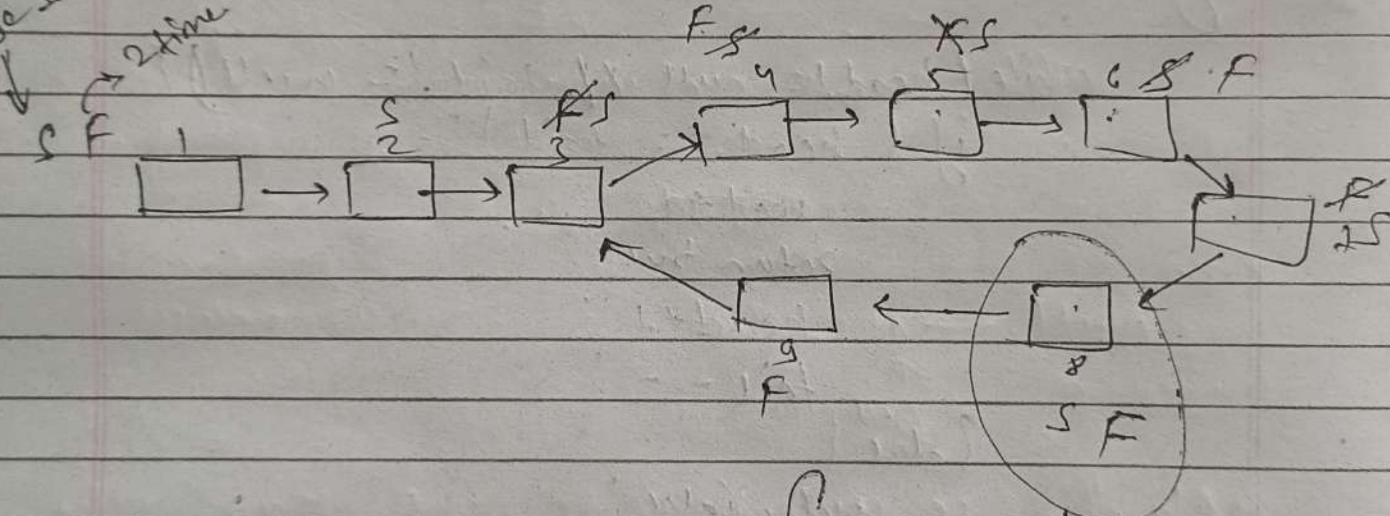
Q4)

head :- [3 → 2 → 0 → -4]



ye 1 time aage badega.

2 time aage badega.



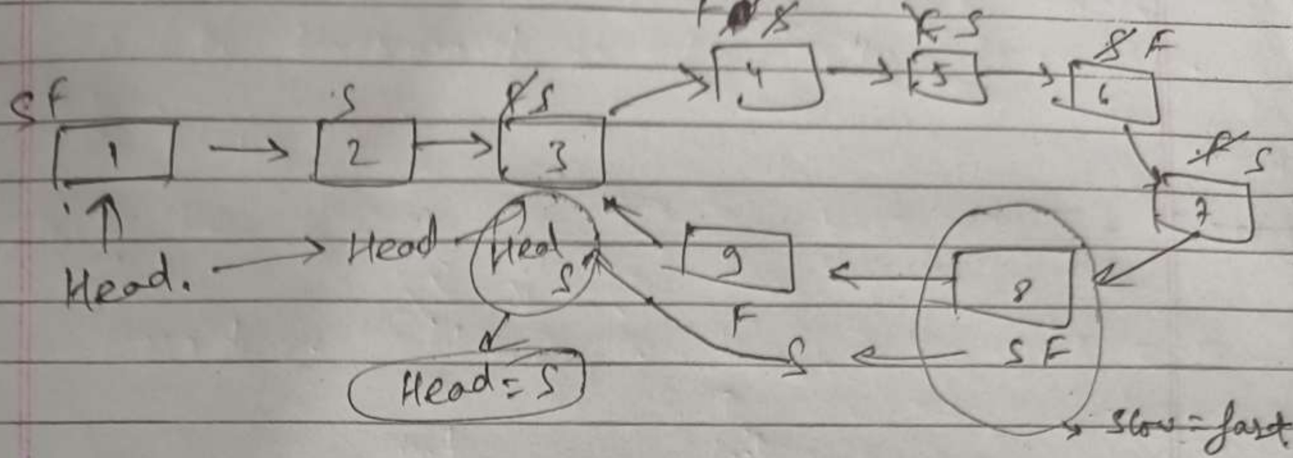
∴ So ye linked list ek cyclic linked list hai.

This is by slow - fast approach.

Is linked list me slow & fast pointer ek hi point pe hai.

cycle detect karne me → Floyd Algorithm use hota hai.

Q5 • cycle kis node se suru hua hai usko pata karne ke liye jis node pe slow = fast hua hai.



• Use node se slow ko 1 step left (or aage) badao aur Head ki taraf se bhi 1 step right ki taraf badao.

• Jaha dono meet karege usi node se cycle start hoga.

Code:-

```
if (fast == slow) {
    ListNode start = Head;

    while (slow != start) {
        start = start.next;
        Slow = slow.next;
    }
    return start
}
return null.
```

Note:- Why does this work?
→ Because distance from head to cycle start equals distance from meeting point to cycle starts.

Note,
Q4 & Q5 same hai